

The dropouts

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Recap

Goal - Identify likely school dropouts.

How? - Classification model - demographic, academic and socio-economic indicators.

So what? - A helping hand for faculty and teachers. Identify early, evaluate how these students can be helped. Provide aid.

Dataset - several disjoint databases, students enrolled in different undergraduate degrees, such as agronomy, design, education, nursing, journalism, management, social service, and technologies

Data Overview

Analysis & Inspection

- Demographics
 - Distributions
 - Risk Assessment
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Sensitive Attributes

- Personally Sensitive Attributes
- Socio-Economic Sensitive Attributes
- Proxies
- Bias, Prejudice & Discrimination

Analysis & Inspection

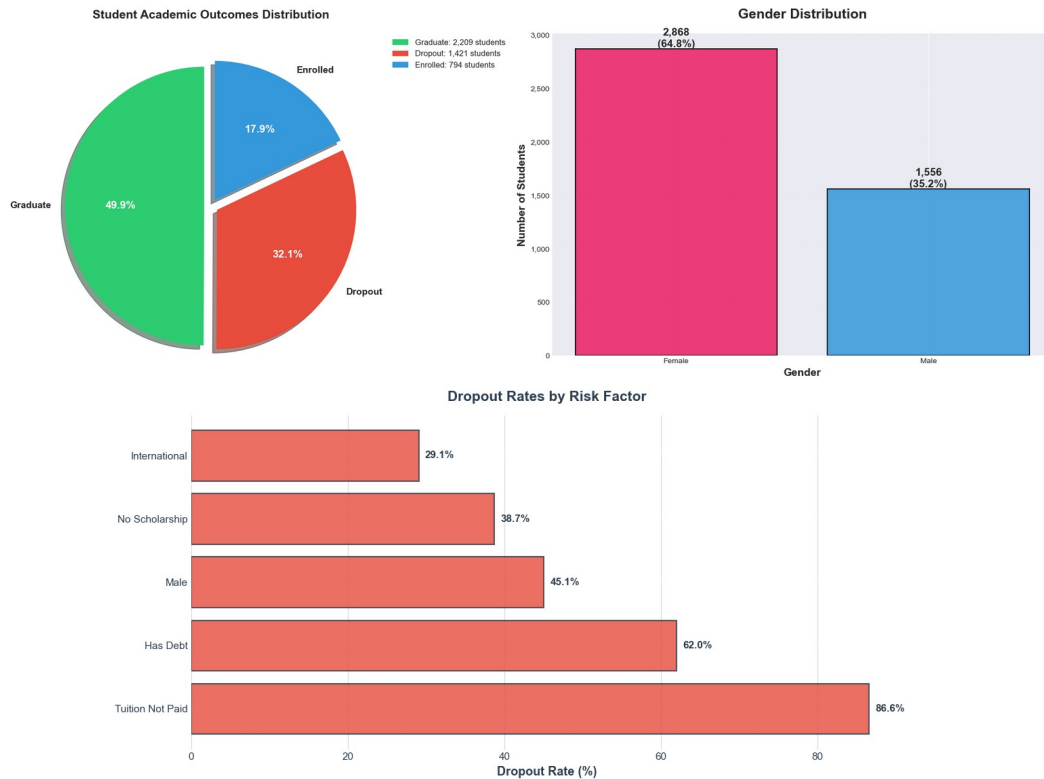
High Dropout Rate (~33%)

Student Background

- Gender Distribution
- Age (median = 20yrs)
 - older → more likely to drop out
- Previous Education
 - Secondary School (84%)
- Nationality
 - Portugal (97.5%) and Portuguese speaking countries

Risk Factor Assessment

- Portuguese Legislation



Sensitive Attributes

Personally Sensitive Attributes

- **Nationality, International Status & Displaced**
 - Proxy for race, ethnicity, migration.
 - **Risk:** xenophobia, bias against minority groups
- **Educational special needs**
 - Proxy for medical/learning disabilities
 - **Risk:** ableist discrimination / privacy violations
- **Marital Status, Gender & Age**
 - Personal family information
 - **Risk:** discrimination against minority groups, sexism & ageism

Socio-Economic Sensitive Attributes

- **Debtor, Tuition Fees up to date & Scholarship Holders**
 - Proxy for financial situation
 - **Risk:** prejudice against lower-income students
- **Mother's and Father's Qualification and occupation**
 - Proxies for social class and family wealth
 - **Risk:** introduces class bias

Pre-processing steps

- **Missing values & outliers already removed**
- **One-hot encoding for categorical variables**
- **Omitting second semester data**
 - Proxy for the target variable
 - Leakage
- **Scaling numerical values**
 - Variables like age & grades
 - Required for machine learning models
- **Omitting currently enrolled students**
 - Binary classification (predicted to successfully graduate or drop out)

Models & Training

- **Preliminary test with different model types**
 - Logistic regression
 - Decision tree
 - Random forest
 - Chosen for best interpretability
- **Stacked classifier**
- **Training process**
 - Cross validation in 5-folds
 - Higher weight for false negatives
- **Training metrics**
 - Accuracy
 - Recall for dropout class
 - Unweighted mean of F1 scores
- **Future work:**
 - Fairness metrics

Results

Model	Accuracy	Dropout recall	F1 score mean
Stacked classifier	0.88	0.85	0.88
Logistic regression	0.89	0.84	0.88
Random forest	0.89	0.79	0.88
Decision tree	0.85	0.78	0.84

- **Stacked classifier only minor improvement**
 - Other model types are more interpretable
- **Best model logistic regression**
 - Suggests high linearity in data
- **Recall is around 85%**
 - ~15% of dropped out students not detected

Evaluation strategy/fairness measurements

→**Metrics:** Primary - Recall + plain ol' FN
(minimize missed kids)

→**ROC curve** to maintain balance

→**Fairness** - Sensitive attributes, what to do?

→**Fairness** - Equal opportunity is our motto
here (More dialogue necessary to understand
how much emphasis we need to put on
Equalized odds and Demographic Parity)

→**Bias mitigation** - pre-processing - checking
imbalance across sensitive groups ‘

Value-sensitive design (DEDA)

Values for different stakeholders:

- **Students** → fairness, privacy, non-stigmatisation
- **Teachers** → usefulness, interpretability
- **University** → wellbeing, transparency, accountability

→ based on values the DEDA process was done

Privacy & legislation:

- Dataset is anonymized, lowering identification risks
- GDPR requires us to be open about how the model uses student data (transparency) and to only use the minimum necessary data for predictions (data minimization)
- Sensitive variables require extra caution
- No judgement based on model output (no discrimination)

Bias & fairness risk:

- Model flag groups more often (family background)
- Check model accuracy across sensitive groups to ensure no subgroup is unfairly treated or misclassified
- Fairness measures can help reduce unequal treatment (weighing, equal opportunity)

Transparency & Explainability

- Teachers must understand *why* a student is flagged
- Students should be informed about the interpretation of the result (advisory)

Conclusion and remaining challenges

Conclusion:

- Model predicts at-risk students well (85% recall)
- Algorithm is used as a supporting tool not for making decisions
- Sensitive attributes require attention
- Fairness measures should be applied to sensitive attributes
- Looking into the laws and regulations that apply to our project
- DEDA highlights importance on privacy, transparency and accountability
- Human judgement stays central -> teachers interpret, not the algorithm

Challenges:

- Risk of flagging vulnerable groups
- Data limitations: data is anonymized but still includes sensitive proxies
- preventing misuse: ensuring teachers interpret the results correctly

